LUCIDMSK CME PROGRAM

MODULE 04

Rheumatologic Arthropathies: Pattern-Based Radiographic Diagnosis

RA · PsA · Gout · CPPD · AS · Sacroiliitis · Six-Axis Framework

1.0 AMA PRA Cat 1 | **General MSK** | **CME Article Review**

LEARNING OBJECTIVES

1. Apply a six-axis framework to systematically analyze arthropathy radiographs.
2. Identify distinguishing features of RA, PsA, gout, CPPD, and AS.
3. Recognize joint-specific patterns: pancarpal RA, pencil-in-cup PsA, overhanging edge gout.
4. Differentiate sacroiliitis patterns: symmetric (AS/IBD) vs. asymmetric (PsA/ReA).
5. Apply modified New York criteria to sacroiliitis grading (0–4).
6. Recognize hallmark deformities of RA and atlantoaxial subluxation implications.

This module is produced by the LucidMSK Editorial Board for educational purposes. Participants completing all content and achieving ≥70% on the post-test may claim the stated AMA PRA Category 1 Credit(s)™. No commercial support received.

1. Six-Axis Framework for Arthropathy Analysis

Axis	Key Question	Discriminating Diagnoses
1. Distribution	Symmetric vs. asymmetric? Ray pattern? Axial?	Symmetric→RA,CPPD; Asymmetric→PsA,Gout; Ray→PsA
2. Bone Density	Periarticular osteopenia? Preserved?	Osteopenia→RA,JIA; Preserved→Gout,PsA,OA,CPPD
3. Erosion	Marginal? Central (seagull)? Overhanging edge?	Marginal→RA; Central→EOA; Overhanging→Gout
4. Joint Space	Uniform narrowing? Focal? Pan-compartmental?	Uniform→RA; Focal→OA; Pancarpal+uniform→RA
5. New Bone	Osteophytes? Syndesmophytes? Periostitis?	Osteophytes→OA; Syndesmophytes→AS; Periostitis→PsA
6. Soft Tissue	Tophi? Calcification type?	Tophi→Gout; Chondrocalcinosis→CPPD; Calcinosis→SSc

2. RA & Psoriatic Arthritis — Key Features

Rheumatoid Arthritis vs. Psoriatic Arthritis

Feature	RA	PsA
Distribution	Symmetric; MCP/PIP/wrist; DIP SPARED	Asymmetric; DIP involved; ray pattern
Bone density	PERIARTICULAR OSTEOPENIA (hallmark)	PRESERVED density — key differentiator
Erosion	Marginal at "bare area"; radial MCP heads first	Pencil-in-cup DIP; "mouse ear"; central erosion
Joint space	Pancarpal collapse; uniform narrowing	Preserved or eccentric; DIP may fuse
New bone	ABSENT (no osteophytes)	Periostitis; enthesophytes; fluffy periosteal reaction
SI joints	Not involved	ASYMMETRIC sacroiliitis
Spine	Step-ladder subaxial; atlantoaxial subluxation	Bulky non-marginal asymmetric ossification

3. Crystal Arthropathies & Sacroiliitis Grading

Crystal Arthropathy Rapid Comparison

Feature	CPPD	HADD (Apatite)	Gout (Urate)
Crystal	Calcium pyrophosphate	Calcium hydroxyapatite	Monosodium urate
Location	TFC (wrist), menisci (knee)	Rotator cuff; peritendinous	1st MTP, ankle, olecranon, wrist
Erosion	Mild; SLAC wrist	None	Overhanging edge; sclerotic margin
Joint space	Narrowed (patellofemoral knee; SLAC wrist)	Preserved	Preserved until late
Special	Crowned dens on CT; chondrocalcinosis	Acute calcific periarthritis	Tophi may calcify late; radiolucent early

Modified New York Criteria — Sacroiliitis Grading

Grade	Finding
0 — Normal	No changes
1 — Suspicious	Equivocal; not clearly sacroiliitis
2 — Minimal	Small erosions or sclerosis; no joint space change
3 — Moderate	Erosion + sclerosis ± widening/narrowing ± partial ankylosis
4 — Ankylosis	Complete bilateral fusion

> **KEY POINT**
>
> ASAS MRI active sacroiliitis: BME on STIR at the SI joint subchondral region in ≥2 consecutive slices, or ≥2 locations on one slice. Gadolinium enhancement is an alternative criterion.

References

- Resnick D, Niwayama G. Diagnosis of Bone and Joint Disorders. 4th ed. Saunders 2002.